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IoT - DNN PROJECT REPORT
BATTERY HEALTH PREDICTION

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VoltX Battery AI: Lithium-Ion Battery Health Prognostics and IoT Digital Twin using Deep Learning

Subject: [Subject Name and Code] | **AAT Project Report** | **Submitted by:** [Your Name], [Register Number] | **Department:** [Department], [College Name] | **Semester:** [Semester] | **Faculty:** [Faculty Name] | **Date:** [Date]

Abstract

Lithium-ion batteries power electric vehicles, portable devices and energy-storage systems, and their capacity fades with every charge and discharge cycle. A Battery Management System (BMS) therefore needs to know not only the present charge but also the State of Health (SOH) and the Remaining Useful Life (RUL) of the cell. This project presents VoltX Battery AI, an intelligent prognostics platform that predicts these quantities from simple measurements of cycle count, discharge voltage, load current and temperature.

A Deep Neural Network (DNN) was trained on the NASA Ames Prognostics Center of Excellence ageing dataset for four 18650 cells (B0005, B0006, B0007 and B0018). The model is served through a FastAPI backend that returns SOH, RUL, estimated capacity, internal resistance, a health grade from A+ to F and a thermal-risk warning. A React dashboard with a glassmorphic dark theme shows the results on a live gauge, forecasts future degradation and keeps a session history. Because physical hardware was not used, an ESP32 with an OLED display, three potentiometers and status LEDs is simulated in Wokwi to act as an IoT digital twin that sends telemetry to the backend over HTTP.

An LSTM model was also tried in the training notebook but produced poor results and was not used for deployment. The work is a working proof of concept, and its accuracy on real cells and real sensors still needs to be validated.

Keywords: battery health, state of health, remaining useful life, NASA dataset, deep neural network, ESP32, Wokwi, FastAPI, digital twin

1. Introduction

1.1 Background

Every charge and discharge cycle slowly damages a lithium-ion cell. Its usable capacity falls, its internal resistance rises and it runs hotter under the same load. When a cell can no longer hold about three quarters of its rated capacity, it is generally considered to have reached end of life. Knowing how close a cell is to that point lets an operator replace it before it fails, avoid unsafe operation and plan maintenance.

Measuring capacity directly needs a full controlled discharge, which is slow and impractical in the field. Data-driven models offer an alternative: they learn the link between easily measured signals and the health of the cell, and they can give an estimate in milliseconds.

1.2 Problem Statement

Simple state-of-charge indicators do not show how worn a battery is, and laboratory capacity tests cannot run during normal use. Rule-based thresholds also miss the gradual, non-linear way a cell degrades. A practical BMS needs an accurate, fast estimate of SOH and RUL from a few routine measurements, an easy way to view the results, and a way to test the whole chain without owning expensive hardware.

1.3 Objectives

1. Train a deep learning model that predicts battery SOH from cycle count, voltage, current and temperature.
2. Derive RUL, capacity, internal resistance, a health grade and thermal-risk warnings from the model output.
3. Serve the model through a documented REST API using FastAPI.
4. Build a React dashboard for prediction, live telemetry, forecasting and history.
5. Simulate an ESP32 hardware node in Wokwi that sends telemetry to the backend and shows status locally.
6. Evaluate the approach honestly and record its limitations.

1.4 Scope

The project covers model training on the NASA dataset, backend inference, the dashboard and the simulated edge device. It does not include tests on physical cells, a real BMS circuit or cloud deployment.

2. Literature Review

Approach	Idea	Limitation
Coulomb counting	Integrates current to track charge	Drifts over time and does not show ageing directly
Equivalent-circuit models	Fit resistance and capacitance parameters	Need careful tuning for each cell type and temperature
Electrochemical models	Simulate internal physics	Accurate but heavy and hard to run on small devices
Data-driven models (neural networks, LSTM, regression)	Learn the mapping from measurements to health	Need good data and can fail outside the range they were trained on

The NASA Ames PCoE dataset is a widely used public benchmark for this task. It contains repeated charge and discharge cycles of 18650 cells until their capacity falls well below the rated value, so the full ageing curve is available. Data-driven prognostics were chosen here because they are light enough to serve in real time and flexible enough to include extra features later.

3. System Analysis and Design

3.1 Architecture

The system has three layers. The hardware layer is a simulated ESP32 node. The AI layer is a FastAPI service that scales the inputs and runs the saved Keras model. The presentation layer is a React dashboard.

Layer	Components	Role
Hardware / IoT	ESP32 DevKit v4 in Wokwi, three potentiometers (voltage, current, temperature), 128 x 64 SSD1306 OLED, green, yellow and red LEDs	Produces sensor-like readings, shows local status, sends JSON over HTTP
AI engine	FastAPI and Uvicorn on port 8000, StandardScaler, TensorFlow / Keras model, prognostics logic	Validates input, predicts SOH and derives all health metrics
Presentation	React and Vite dashboard on port 5173, SVG gauge, Recharts	Prediction, telemetry stream, forecast and history views

Data flows from the ESP32 (or from the dashboard sliders) as a JSON request to the backend. The backend applies the saved scaler, runs the model and returns the full set of prognostic metrics, which the dashboard displays.

3.2 Technology Stack

Area	Technology
Machine learning	Python, TensorFlow and Keras, scikit-learn (StandardScaler), Pandas, NumPy, Matplotlib, Google Colab
Back end	FastAPI, Uvicorn, Swagger documentation at `/docs`
Front end	React 19, Vite, Recharts, custom SVG gauge, glassmorphic dark theme
IoT simulation	Wokwi, ESP32 DevKit v4, Arduino-style C++ sketch with WiFi and HTTP client

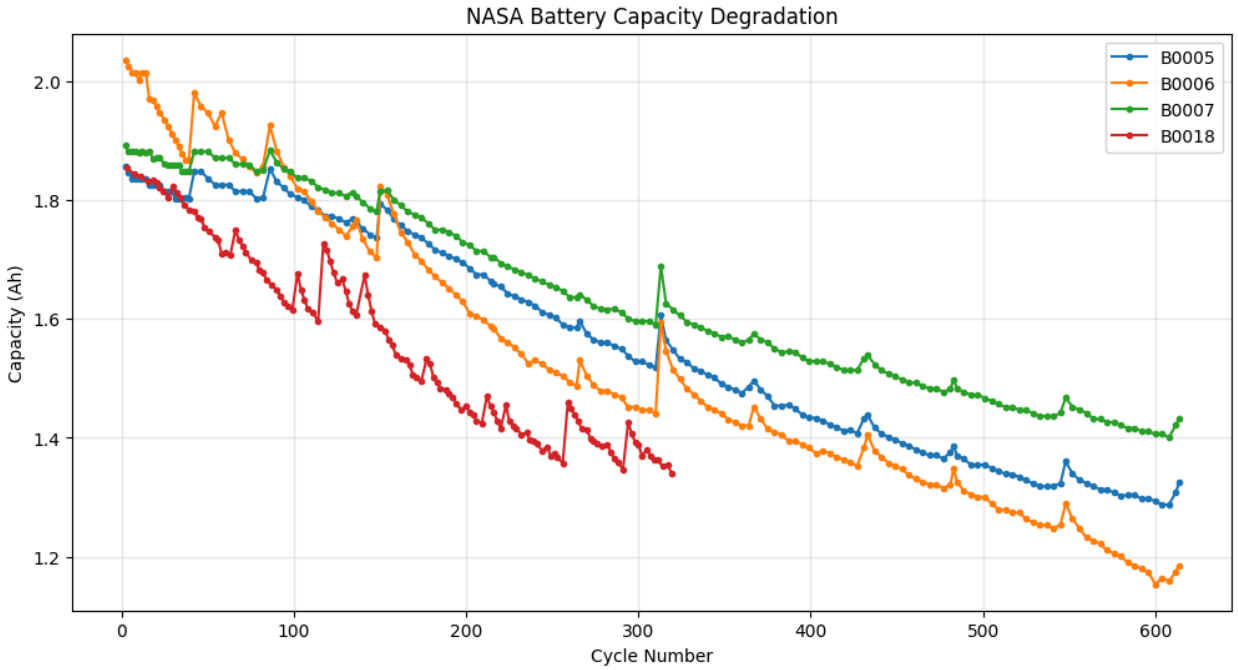
4. Methodology

4.1 Dataset

The training data comes from the NASA Ames Prognostics Center of Excellence 18650 lithium-ion ageing dataset. After preprocessing, the table has 636 rows and 17 columns, one row per discharge cycle, drawn from four batteries.

Battery	Cycles (rows)
B0005	168
B0006	168
B0007	168
B0018	132
Total	636

Each battery is split in time order, not randomly. For B0005 and B0006, cycles 1 to 134 (134 rows) are used for training and cycles 135 to 168 (34 rows) for testing. A chronological split is the honest choice for prognostics, because the model is tested on later, more degraded cycles that it has never seen, just as it would be in real use.



The capacity plot shows the expected ageing behaviour. All four cells start between roughly 1.85 and 2.03 Ah and decline steadily, with B0018 falling fastest and B0007 slowest. Small upward spikes appear at intervals, which are capacity recovery effects after rest periods and not real healing of the cell. These spikes add noise that the model has to tolerate.

4.2 Input Features and Scaling

The model uses four inputs.

Feature	Meaning
Cycle	Number of charge and discharge cycles completed

Feature	Meaning
Voltage (V)	Mean discharge voltage
Current (A)	Mean load current (negative means discharge)
Temperature (°C)	Mean discharge temperature of the cell

A StandardScaler is fitted on the training data so that every feature has zero mean and unit variance. The same scaler is saved and applied in the backend, which keeps the training and inference pipelines identical.

4.3 DNN Architecture

Layer	Configuration
Input	4 scaled features
Dense 1	64 neurons with Batch Normalisation and Dropout 0.15
Dense 2	32 neurons with Dropout 0.10
Dense 3	16 neurons
Output	1 neuron predicting SOH

Batch normalisation stabilises training, and light dropout reduces over-fitting on the small dataset. The network is small on purpose, because only 636 samples are available and a larger model would simply memorise them.

4.4 Health Metrics Derived from the Prediction

The backend turns the model output into the quantities a BMS user needs.

- **State of Health (SOH %)** is the remaining fraction of rated capacity: SOH = capacity ÷ rated capacity × 100. The rated capacity of the 18650 cell is 2.0 Ah.
- **Remaining Useful Life (RUL)** is the number of cycles projected until SOH falls to the 75% end-of-life threshold, which corresponds to 1.5 Ah.
- **Capacity (Ah)** is the effective charge the cell can hold.
- **Internal resistance (mΩ)** is an estimated electrochemical resistance that rises as the cell ages.
- **Health grade** classifies the cell from A+ to F, and a **thermal-risk** flag warns when the temperature is elevated.

4.5 LSTM Experiment

The notebook also trains an LSTM model on the same data. Its test results were MAE 68.44, RMSE 68.76 and R^2 of −117.88. These values are far outside any sensible range for a quantity that lies between 0 and 100 and show that the LSTM run failed. The most likely causes are a target or feature scaling mismatch or an input-shape problem, though this was not investigated here. The LSTM was therefore not used, and the DNN was chosen for deployment.

5. Implementation

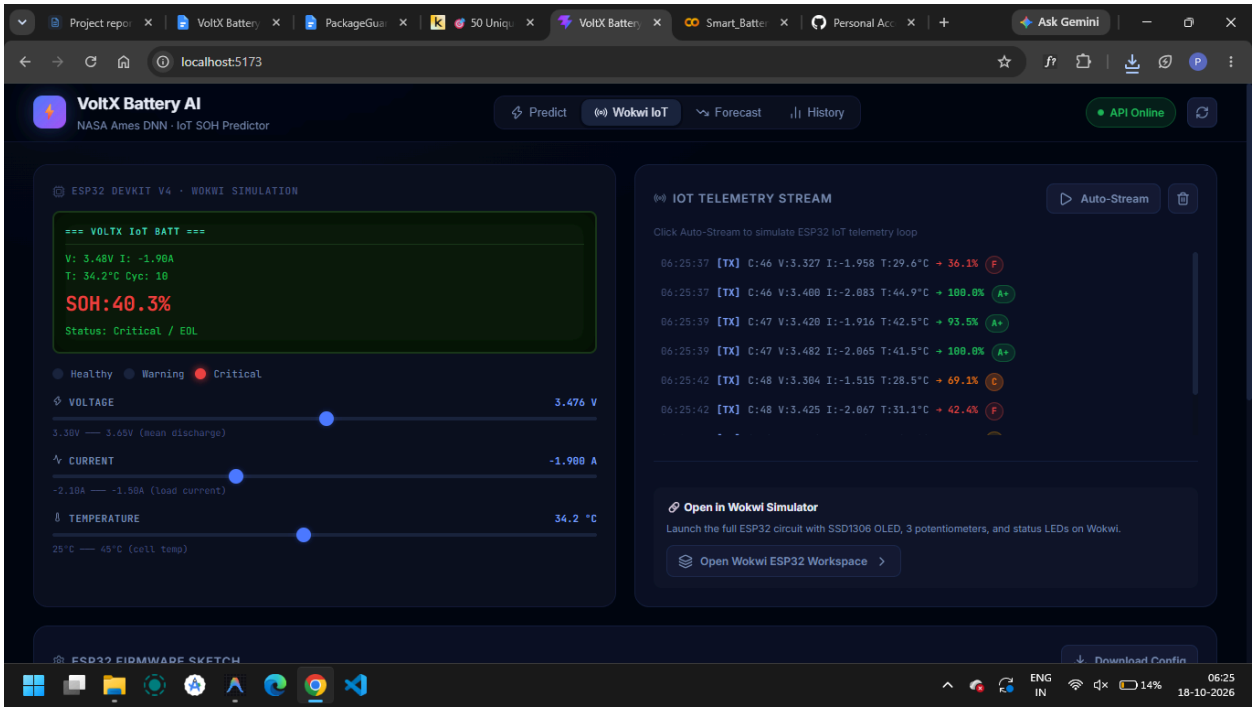
5.1 FastAPI Backend

The backend (`app.py`) loads the scaler and the Keras model at start-up, validates each request, runs inference and returns SOH, RUL, capacity, internal resistance, health grade and thermal-risk text in one JSON response. Swagger documentation is available at `http://127.0.0.1:8000/docs`, and CORS is enabled so the dashboard on port 5173 can call it. Run it with `uvicorn app:app --reload --port 8000`.

5.2 ESP32 Digital Twin in Wokwi

The circuit uses an ESP32 DevKit v4 with three potentiometers on ADC pins 32, 34 and 35 to mimic voltage, current and temperature, a 128 x 64 SSD1306 OLED on I2C, and three LEDs. The firmware connects to WiFi, reads the potentiometers, posts the readings as JSON to the backend, and shows the voltage, current, temperature, cycle count, SOH and status on the OLED. The LEDs show the health state: green for healthy, yellow for warning and red for critical.

5.3 React Dashboard



Tab	Purpose
Predict	Four sensor inputs, an instant AI evaluation, a live SOH gauge, four summary cards (SOH, RUL, cell capacity, internal resistance) and engineering recommendations

Tab	Purpose
Wokwi IoT	A virtual ESP32 panel with OLED mock-up, status LEDs and sliders for voltage, current and temperature, an auto-stream control for the telemetry loop and the firmware sketch with a download option
Forecast	Forward trajectory of SOH across 20 to 100 future cycles, drawn with Recharts
History	A session table of diagnostics with an export option

A status badge shows whether the API is online, and the interface is responsive and uses a dark glassmorphic design.

6. Results and Discussion

6.1 Predictions Across Battery Life

The dashboard presets reproduce three stages of cell life, and the results are shown below.

Metric	Cycle 10 (fresh)	Cycle 60 (used)	Cycle 100 (degraded)
Discharge voltage	3.56 V	3.48 V	3.42 V
Discharge current	−1.82 A	−1.89 A	−1.95 A
Temperature	31.5 °C	33.4 °C	35.2 °C
Predicted SOH	100.0%	91.4%	82.26%
Health grade	A+	A	B
RUL estimate	more than 200 cycles	about 115 cycles	40 cycles
Internal resistance	18.6 mΩ	22.4 mΩ	30.4 mΩ
System status	Optimal condition	Normal ageing	Secondary degradation detected

The trend is physically sensible: as the cell ages, voltage falls, current draw and temperature rise, SOH drops, RUL shrinks and internal resistance climbs.

6.2 Model Accuracy

The DNN test metrics should be stated here from the notebook: MAE [value], RMSE [value] and R^2 [value] on the held-out cycles, together with a plot of actual versus predicted SOH.

```
LSTM RESULTS
-------------------------
MAE : 69.44395641215863
RMSE: 69.75956729655955
R2  : -113.04654205181835
```

6.3 Behaviour Outside the Training Range

Typing a different combination into the dashboard shows the limits of the model. With cycle 10, 3.476 V, −1.9 A and 34.2 °C, the system reports SOH 40.3%, RUL 0 cycles, capacity 0.807 Ah, internal resistance 57.6 mΩ and grade F, and the OLED in the Wokwi panel shows the same critical status. A cycle count of 10 normally means a nearly new cell, so this result is inconsistent with the cycle number. It also falls outside the training data, because the NASA cells never dropped below about 1.15 Ah while this output implies 0.807 Ah.

This shows that the model relies mainly on voltage and temperature and does not enforce a physical link to cycle count. A neural network extrapolates unpredictably beyond the data it has seen, which is why range checks and consistency checks are needed in a real BMS.

6.4 Discussion

The project shows that a small DNN on four features can drive a full prognostics pipeline, including a simulated hardware node and a usable dashboard. The chronological train and test split and the use of public data make the approach credible. The main weaknesses are the small dataset of 636 rows, the failed LSTM experiment and the sensitivity to inputs outside the training range. Derived values such as RUL, resistance and grade depend on the SOH estimate, so any error in SOH carries through to all of them.

7. Limitations

1. The data covers four cells of one type under laboratory conditions, so results may not carry over to other chemistries or real driving loads.
2. The model can give unrealistic outputs for inputs outside the training range, as the cycle 10 example shows.
3. The LSTM experiment failed and has not been fixed.
4. The ESP32 is simulated, and the potentiometers are not real battery measurements.
5. Internal resistance and RUL are derived estimates and were not checked against measured values.
6. The final DNN error metrics still need to be reported from the notebook.

8. Conclusion and Future Scope

VoltX Battery AI combines the NASA battery ageing dataset, a compact DNN, a FastAPI service, a React dashboard and a Wokwi ESP32 digital twin into one end-to-end battery health monitor. It estimates SOH, RUL, capacity and resistance in real time, grades the cell, warns about temperature and forecasts future degradation, and the whole chain can be demonstrated without physical hardware.

The project is a solid proof of concept. Its next steps are to report full accuracy metrics, correct the behaviour outside the training range and test on real cells.

Future enhancements

- Fix the LSTM pipeline and compare it fairly against the DNN using the same chronological split.
- Add input range checks and physical-consistency rules, such as limiting SOH by cycle count.

- Train on more cells and conditions, including other NASA batteries and additional public datasets.
- Add uncertainty estimates so the dashboard shows a confidence range for SOH and RUL.
- Replace the simulation with a real ESP32, current and voltage sensors and a temperature probe.
- Store history in a database and add alerts by email or mobile notification.
- Deploy the backend to the cloud and convert the model to a lightweight format for edge inference.

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PROJECT SOURCE CODE LINK :

https://github.com/kit28cs119/battery